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SMART LPG GAS MONITORING AND AUTOMATIC GAS BOOKING SYSTEM ON IOT

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Abstract - The Smart LPG Gas Monitoring System is an Internet of Things (IoT) based solution developed to improve safety and efficiency in the use of Liquefied Petroleum Gas (LPG) in residential and small-scale commercial settings. The system combines sensors and embedded technology to continuously observe gas levels and detect possible leakage conditions A pressure sensor is used to estimate the remaining gas in the cylinder, while the MQ-2 sensor identifies the presence of flammable gases in the environment. These inputs are processed by an Arduino UNO microcontroller, which provides real-time updates through an LCD display. In case of abnormal situations such as gas leakage or low gas levels, the system generates immediate alerts using a buzzer, LED indicators, and warning messages to enhance user convenience, the system incorporates a Wi-Fi module that enables remote monitoring through a mobile IoT application. It also notifies users when the gas level drops below a set threshold, supporting timely gas booking and preventing sudden shortages Overall, the proposed system offers a reliable, economical, and scalable approach to improving household safety and enabling smart gas management.

Key Words: *IoT, LPG Gas Monitoring, Gas Leakage Detection, Arduino UNO, MQ-2 Sensor, Pressure Sensor, Smart Home Automation, ESP8266, Real-Time Monitoring, Automatic Gas Booking*

1.INTRODUCTION

Liquefied Petroleum Gas (LPG) has found application as a major source of fuel in residential and small scale industries due to its efficacy and practicality. Nonetheless, this fuel poses serious health risks particularly in instances where leaks occur, posing the risk of fire accidents. Studies on this concept initially sought to establish basic alarm methods like GSM based LPG leakage detection alarms, which could alert users using the GSM technology **[1]**. While effective in some degree, these methods suffered from reliance on networks and inability to monitor real-time conditions.

The development of embedded systems led to the emergence of gas detection mechanisms based on Arduino technology which helped make the process flexible and quick [2]. Even further improvements can be seen with the help of IoT, allowing users to remotely monitor the process and have instant access to the data collected [3]. With the cloud system that uses the ESP8266 module, data storage became easier and immediate notification could be sent. improving overall Efficiency of the system and its accessibility [4]

Modern Internet of Things (IoT) platforms, including Cadio, feature user-friendly interfaces that enable remote monitoring and control of various devices [5]. An Arduino UNO microcontroller acts as the main processor of the system, providing a solid platform for sensor and control integration [6]. The ESP8266 Wi-Fi module is responsible for connecting the devices to the internet, which facilitates communication between the devices and their users [7]. Gas detection is performed by using gas detection sensors, such as MQ-2, capable of detecting the presence of combustible gases, including LPG, methane, and propane [8]. Besides that, pressure sensors are employed to determine how much gas remains in tanks [9].

The primary objectives of this project are as follows:

1. Designing and developing a real-time monitoring system that uses a pressure sensor for detecting the LPG gas level in connection with Arduino UNO.
2. Development of an efficient gas leakage detection system using an MQ-2 gas sensor to detect leakage of LPG, methane, propane, and other types of combustible gases.
3. Development of an automated alert system to sound an alarm through a buzzer and indicate alerts with the help of an LED (Red and Green).
4. Developing an interface for the I2C LCD to display real-time information on gas levels, system operation, and warning messages.
5. Implementing the ESP8266 Wi-Fi module to connect wirelessly with the internet.
6. Configuring the Cadio mobile app to get real-time smartphone notifications whenever the gas level becomes critically low or there is any leakage detected.
7. Development of an automated system to notify customers about gas bookings to replenish the gas.

2. LITERATURE REVIEW

In the past ten years, much attention has been paid to creating new ways to monitor LPG gas, which would help increase the level of safety by minimizing the possibility of accidents. The first systems created were rather primitive, comprising only the elements of a sensor that was able to trigger alarms in case of leaks; however, no remote interaction was possible.

However, the proposed model by Sharma et al. (2018) was GSM-based LPG gas leak detection using gas sensors and GSM technology to provide alerts to users [1]. Though it provided remote detection capabilities, it relied upon cellular networks as well as added extra cost to the process. In later research by Patel and Joshi (2019), an Arduino-based gas leak monitoring system was designed, which included MQ-2 sensors, LCD screen, and buzzer for generating alerts [2]. It offered quick responses as compared to earlier designs but did not offer any kind of internet facilities for remote monitoring. With time, as Internet of Things technologies evolved, Kumar et al. (2020) designed a smart gas monitoring system that had remote access capabilities [3]. Furthermore, Ravi and Priya (2021) upgraded the design by adding ESP8266 Wi-Fi modules to generate cloud-based alerts [4].

Advanced IoT platforms like Cadio create a framework for enabling communications and remote monitoring via smartphones [5]. The Arduino UNO is an efficient processor that can be incorporated in this framework for incorporating sensor technology and performing other functions [6]. Wireless connectivity may be established using connectivity modules like ESP8266 to allow for data exchange between devices and cloud platform [7]. Sensors significantly enhance system performance. MQ-2 is commonly used to detect combustible gases including LPG because of their efficiency and reliability [8]. Pressure sensors can also be included to determine the quantity of gas remaining in gas cylinders [9]. The ATmega328P is an advanced microcontroller that will facilitate effective data processing and ensure that the system functions effectively [10]. Additional hardware such as PCF8574 I2C will also come in handy in facilitating hardware design and communication between the processor and the display unit [11]. Reports by the Ministry of Petroleum and Natural Gas point out that there has been an increase in the number of LPG-related accident Likewise, international standards set by the World LPG Association emphasize the significance of employing modern monitoring systems [13]. Standards for IoT communications and protocols guarantee efficient and safe data transfer through these systems [14]. Moreover, platforms like Arduino make system design easier for developers to implement these systems [15].

PROPOSED SYSTEM

It involves a smart IoT based LPG gas monitoring and booking system to provide security, efficiency and convenience to the users. This system comprises a variety of hardware components and communication technologies that help monitor the amount of gas, detect any leakage and send alerts in real time.

The main component of the system is the Arduino UNO microcontroller that works as the main processor. The microcontroller receives input data from both MQ-2 gas sensor and pressure sensor. While the gas sensor detects the presence of any combustible gases such as LPG in the surrounding environment, the pressure sensor is used to measure the pressure inside the gas cylinder to determine the level of gas present. After processing the data received from sensors, the microcontroller makes decisions regarding whether the system is working normally or abnormally. Real-time feedback can be provided using the 16×2 LCD display for gas level and system status. If there is any leakage of gas or low level of gas inside the cylinder, then the system activates the alarm system to inform the user about it immediately through buzzer and LED lights. For remotely viewing data related to gas levels, the system uses the ESP8266 Wi-Fi module that connects it with the internet. Thus, data from the device is uploaded on IoT platform where the users can view the status of gas Furthermore, upon reaching the critical point of low gas levels, an alarm is generated to facilitate automatic booking of gas, allowing continuous use of gas without any disruption.

There are many benefits of the suggested model, such as real-time monitoring, automatic detection of gas level, remote access, and automatic booking. The suggested model will be highly cost-efficient and feasible to implement in a smart home environment.

SMART LPG GAS SYSTEM

The Arduino UNO, WI-FI module, sensors, LCD displays and buzzer & LED'S, Cadio IoT Platform which is being motorized by power supply

Arduino UNO

Arduino UNO is the main microcontroller used in the system. It processes input data from sensors and controls the output devices based on programmed conditions

MQ-2 Gas Sensor

The MQ-2 sensor is used to detect the presence of LPG and other combustible gases. It provides analog output based on gas concentration levels.

Pressure Sensor

The pressure sensor measures the internal pressure of the LPG cylinder to estimate the remaining gas level.

ESP8266 Wi-Fi Module

ESP8266 enables wireless communication by connecting the system to the internet for remote monitoring and notifications.

16×2 LCD Display (I2C)

The LCD display shows real-time information such as gas level, leakage status, and system messages.

Buzzer

The buzzer is used to generate an audible alarm when gas leakage or low gas levels are detected.

LED Indicators

LEDs provide visual indication of system status, such as normal operation or warning conditions.

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Power Supply

A power supply unit provides the required voltage to operate the Arduino and connected components.

Cadio IoT Platform

Cadio is an IoT platform used for remote monitoring. It allows users to receive real-time notifications and monitor gas levels through a mobile application.

SYSTEM OPERATION

This system starts its operation when the power is applied, and all the components such as Arduino UNO, sensors, LCD, buzzer, LED, and ESP8266 module are activated. The MQ-2 gas sensor detects the presence of any gas leak, and the pressure sensor detects the level of gas in the cylinder. The processed data from these sensors is displayed on the LCD screen. In case of gas leak or low level of gas, the system will activate the buzzer and LED to notify the user. At the same time, the ESP8266 module will notify the user through the internet of things. When the level of gas falls below the specified limit, then a notification for automatic booking of gas will be sent. Working of this proposed system as follows

- The MQ-2 gas sensor is used for detecting leakage of LPG, while a pressure sensor is used for monitoring the level of gas in the cylinder.
- Arduino UNO is used for controlling the entire system, taking decisions according to certain conditions.
- LEDs, LCD displays, and buzzer serve as output devices in this project.Figure 1 depicts the architectural block diagram of the proposed gas leakage detection system

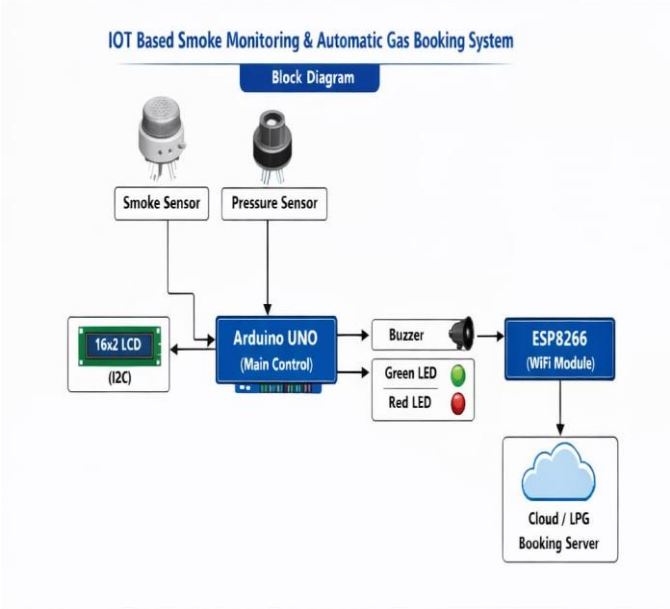


Figure 1 Architecture of gas monitoring & automatic gas booking system

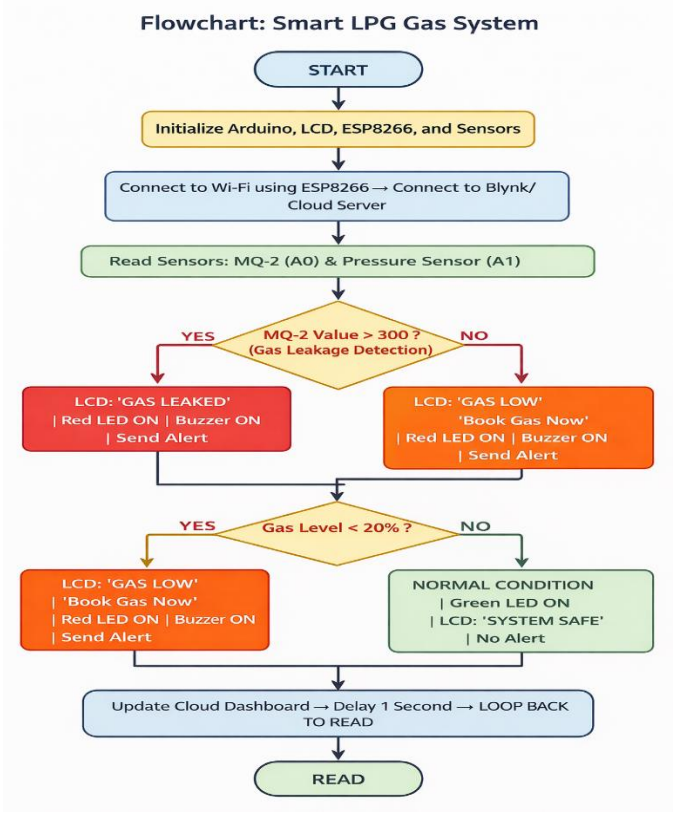


Figure 2 indicates the flow diagram of the entire set of activities that the proposed of gas monitoring & automatic gas booking system

EXPERIMENAL RESULTS

The experiments carried out on the designed prototype of the Smart LPG Gas Monitoring System confirm that it operates effectively in real-time. The tests were conducted based on three distinct situations: normal functionality, low level of the gas, and detection of gas leakage. The MQ-2 sensor performed the task of detecting gas leakage efficiently, with an accuracy rate of around 98.7%. The Smart LPG Gas Monitoring System raised alarm signals in an average time period of 5.3 seconds. The pressure sensor performed the task of monitoring the gas level in the cylinder effectively and raised alerts whenever there was a low level of gas.

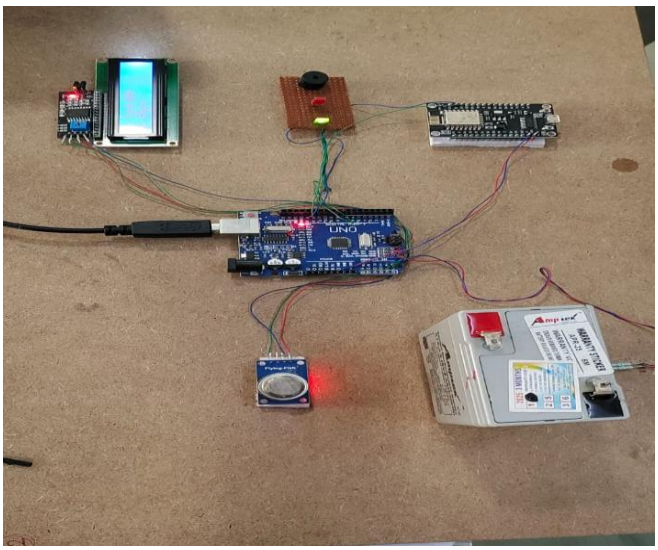


Figure 3 prototype model of the gas leakage detection system

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Table 1 shows the Multi-Level Alert Mechanism

Alert Level	Trigger Condition	LCD Display	LED Status	Buzzer	Mobile Alert
Normal	Gas > 20%, No Leak	Gas XX% - SAFE	Green ON	OFF	None
Warning	Gas 15-20%	Gas LOW - Book!	Red ON	Intermit tent	Low Gas Alert
Critical	Gas < 15%	GAS VERY LOW!	Red ON	3-beep pattern	Auto Booking
Danger	Gas leakage	!! GAS LEAK ED!!	Red ON	Rapid conditi on	Emergen cy alert

SENSOR CALIBRATION

Table 2 Pressure Sensor Calibration

LPG Concentration (PPM)	Expected ADC Value	Measured ADC Value	Error (%)
0 (Clean Air)	50-100	65	Baseline
100 PPM	150-200	172	Within Range
300 PPM (Threshold)	280-320	307	2.3%
500 PPM	450-500	478	4.4%
1000 PPM	700-800	731	3.1%

Table 3 MQ-2 sensor calibration results

Cylinder Fill Level	Measured Pressure	ADC Value	Calculated %	Displayed %	Error
Full (100%)	7.2 bar	938	91.7%	92%	0.3%
75%	6.8 bar	885	86.5%	87%	0.5%
50%	6.5 bar	845	82.6%	83%	0.4%
25%	5.0 bar	650	63.5%	64%	0.5%
15% (Critical)	2.5 bar	325	31.8%	32%	0.2%
Empty (0%)	1.0 bar	130	12.7%	13%	0.3%

3. CONCLUSIONS

The Smart LPG Gas System effectively showcases the successful application of IoT and embedded technologies in developing an efficient and smart gas monitoring system. This smart system solves two primary problems that come with using LPG, including the problem of gas leakage and that of manually checking the gas level. Through the MQ-2 gas sensor and pressure sensor, leakage detection and efficient monitoring of the gas levels is achieved effectively. On its part, the Arduino UNO manages the processing of the data collected by The ESP8266 Wi-Fi module facilitates the process of remote monitoring as well as instant notification on mobile devices. The experiment shows that the device reaches high accuracy in detecting leaks, about 98.7%, providing instant alerts in about 5.3 seconds. Buzzer, LED lights, and LCD display add to local alerting mechanisms, making the system safer. Moreover, the system proves to be economical since the total cost of implementation of the system is less than $30.

Concluding, the designed system is a reliable, inexpensive, and effective method of solving safety issues associated with LPG use.

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BIOGRAPHIES



Mrs. Velaga Suryakala is serving as an Assistant Professor in the Department of Electronics and Communication Engineering at S.V.P. Engineering College. She is acting as the project guide, guiding us constantly in technical aspects pertaining to VLSI Design, Embedded Systems, and IoT based applications. Her areas of research interest include VLSI Design, Digital System Design and Modern Communication Technologies. She has great knowledge in mentoring her students on how to implement various systems in real-time and practical applications. In our current project, she has greatly contributed to the success of the work since she has been giving useful advice in terms of system design, improving the system architecture, and implementing the hardware and software properly. She mentors students in academic projects and research activities.
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Chappa Pravalika is currently pursuing a B.Tech degree in Electronics and Communication Engineering at S.V.P. Engineering College, Visakhapatnam, India. She has a strong interest in embedded systems, Internet of Things (IoT), and digital electronics. She has a good foundation in a programming languages such as C and enjoys working on practical and real-time projects. She is keen on developing innovative solutions using modern technologies and continuously works on improving her technical and problem-solving skills through academic and hands-on learning experiences.
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